EECT 6325 – VLSI Design
Project 3: Inverter Design and Layout

Adil Bhojwani (axb210168)
Corey Finch (cxf210015)
Paresh Grover (pxg210048)
Date: 10/16/2026

Measured Results
Simulated on the PEX-extracted INVTAP netlist, VDD = 0.8 V, 20 fF load, 20 ps input slew

Rise delay (tLH): 45.7 ps
Fall delay (tHL): 44.1 ps
Output rise edge rate (20-80%): 49.4 ps
Output fall edge rate (80-20%): 46.6 ps
Layout area (M1 to M1, INV cell): W × H = 0.36 μm × 0.666 μm = 0.240 μm^2

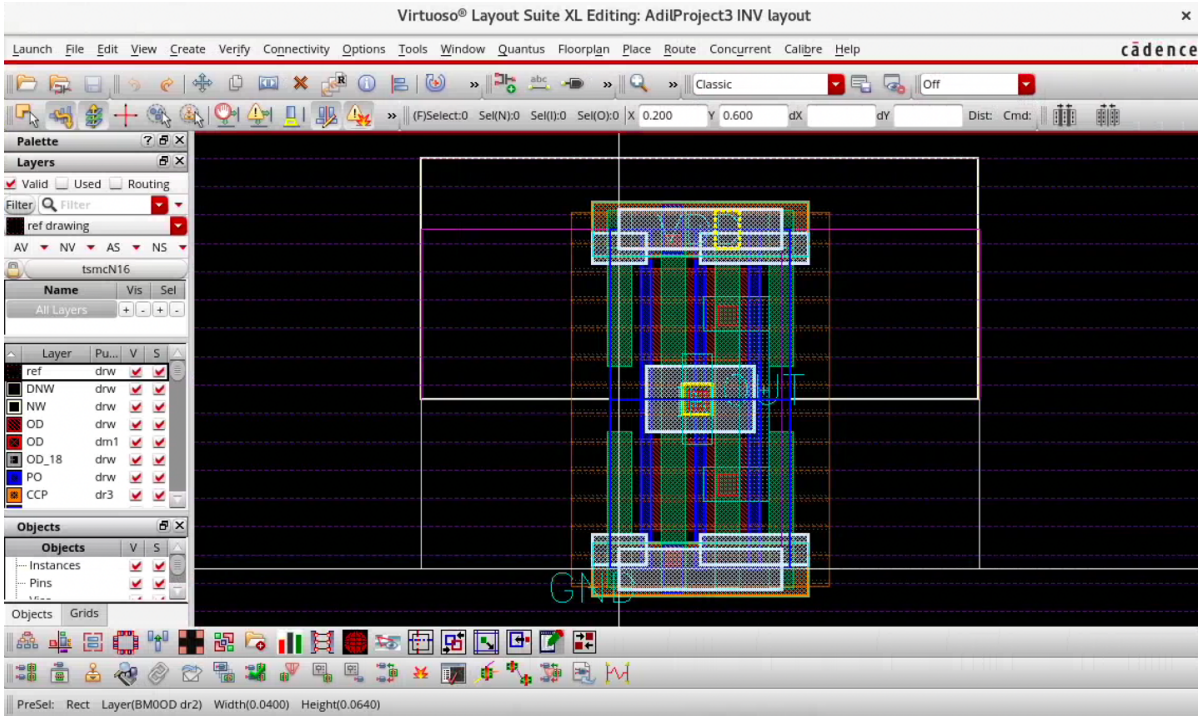


Figure 1: INV layout (TSMC 16nm FinFET, Cadence Virtuoso). Single-gate inverter cell with 4-fin nMOS and pMOS devices, M0OD/M0PO/VIA0 connections, and M1 VDD and GND rails. IN and OUT pins are labeled on M1.

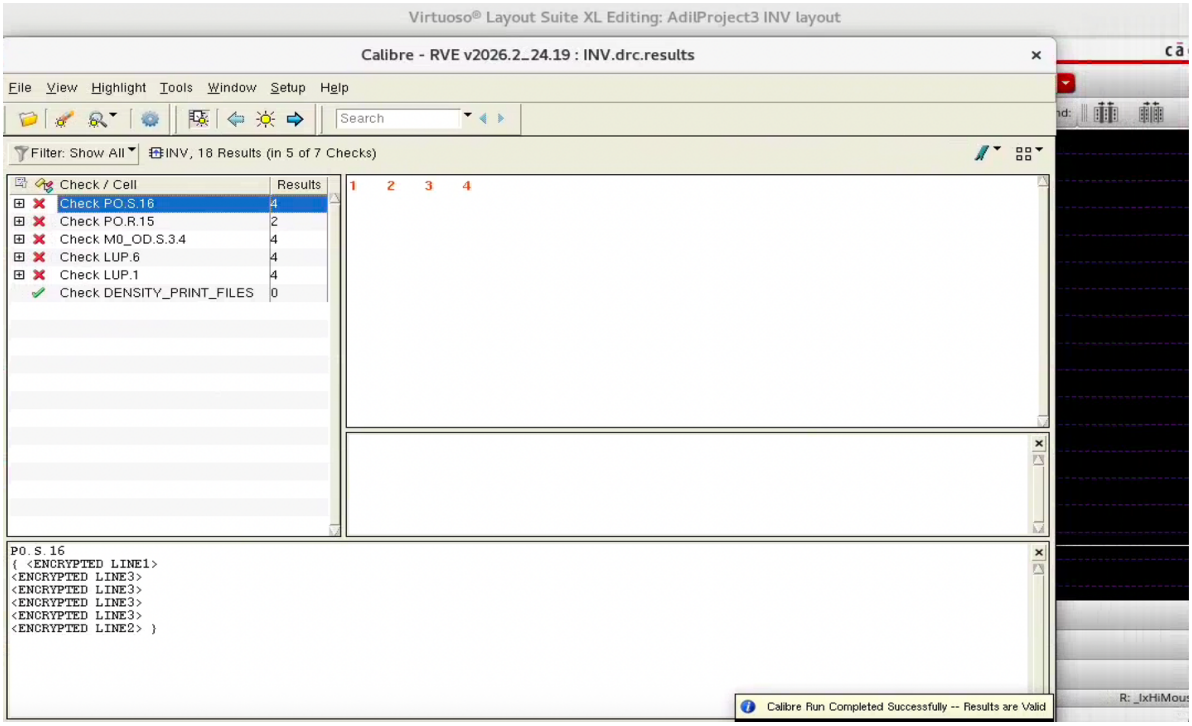


Figure 2: DRC results for the INV cell. Calibre nmDRC reports the five expected violations for a standalone cell without a tap (PO.S.16, PO.R.15, M0_OD.S.3.4, LUP.6, LUP.1). These match the tutorial's expected results and are resolved or replaced in the INVTAP cell.

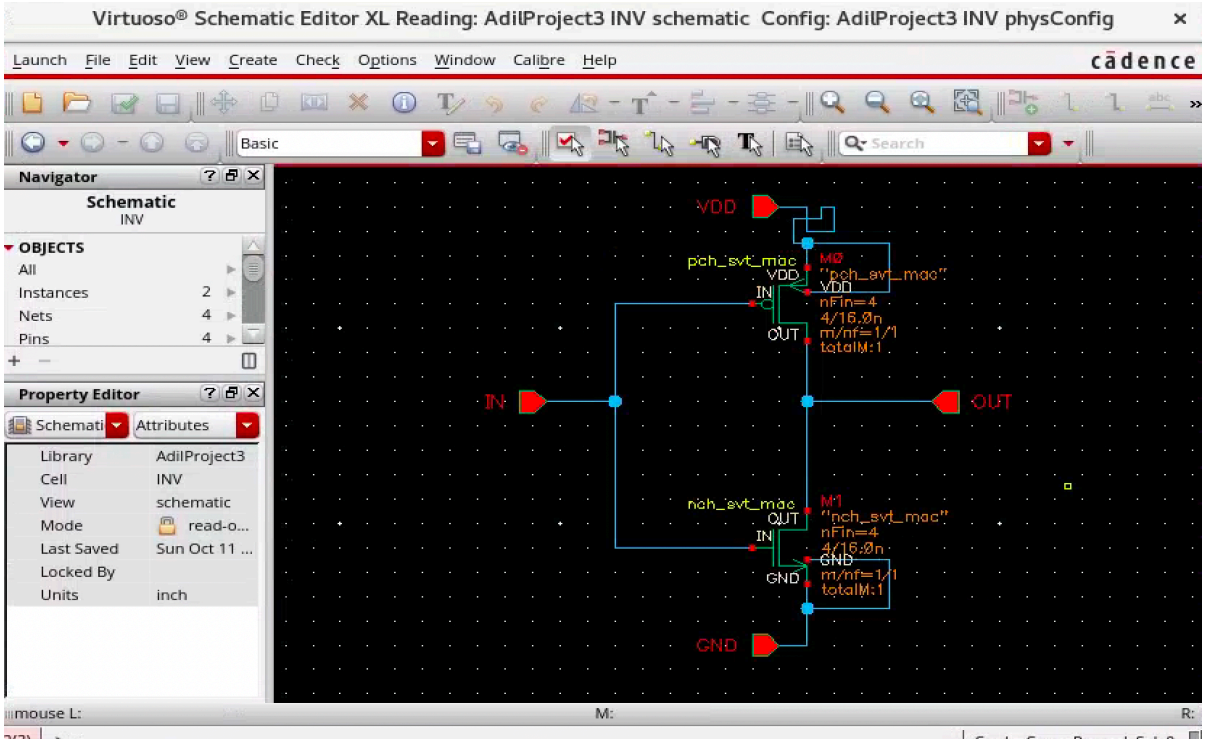


Figure 3: INV schematic. pch_svt_mac (M0) and nch_svt_mac (M1), each with nfin=4 and L=16nm, with IN, OUT, VDD, and GND pins. Used as the source netlist for LVS.

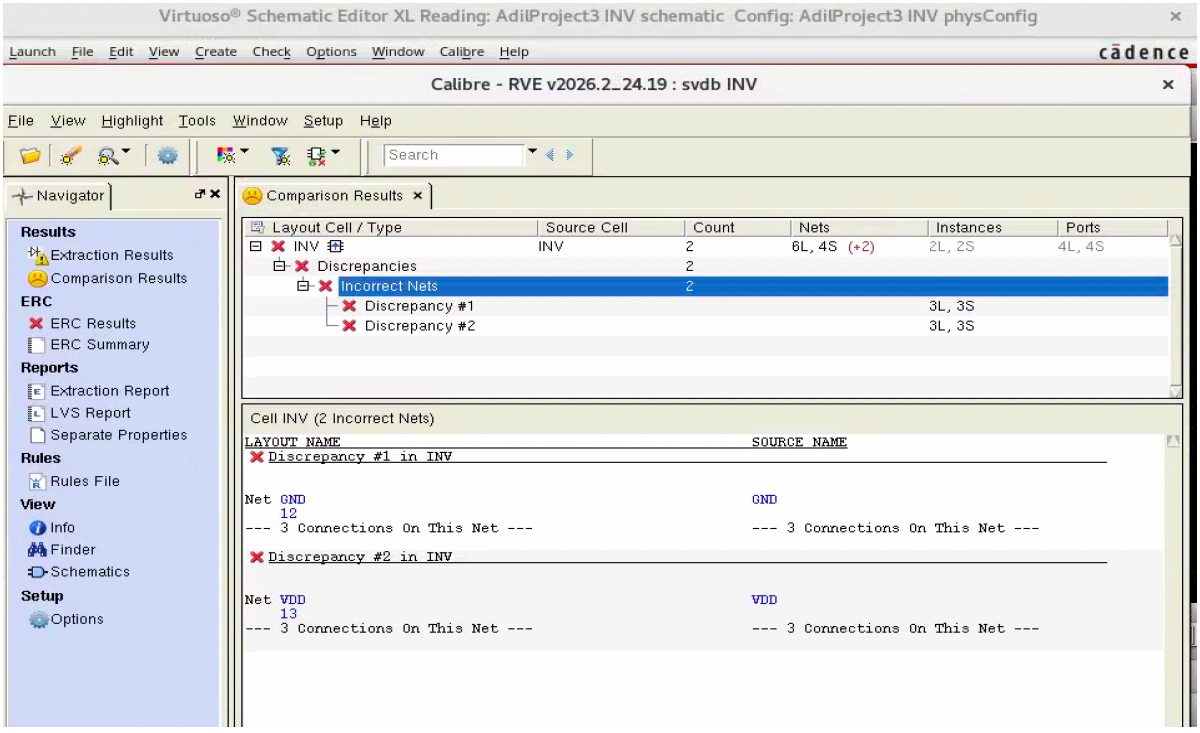


Figure 4: LVS results for the INV cell (pre-tap). Calibre nmLVS reports two incorrect nets: VDD and GND each show three layout connections versus two in the source. This is expected because the standalone INV has no source-to-bulk tie.

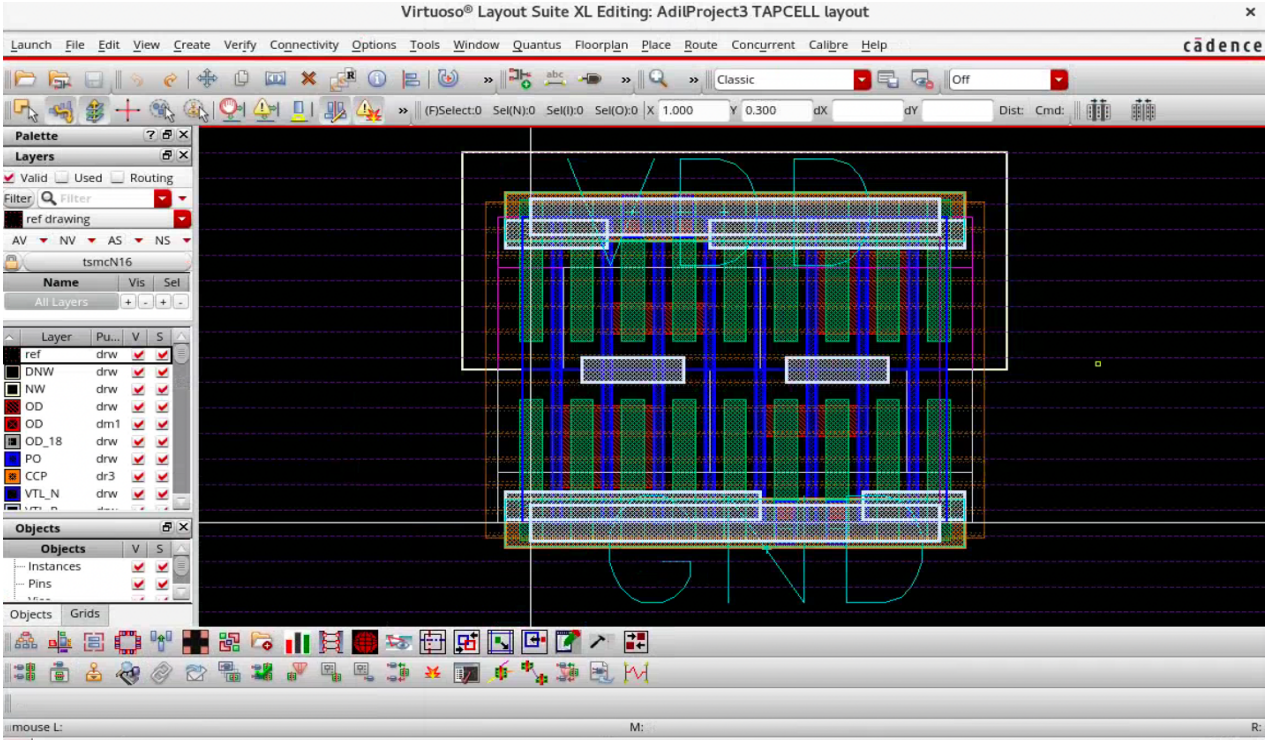


Figure 5: TAPCELL layout. Seven-poly tap cell providing the N-well and P-substrate contacts, with PP and NP implant pockets defining the source-bulk connections.

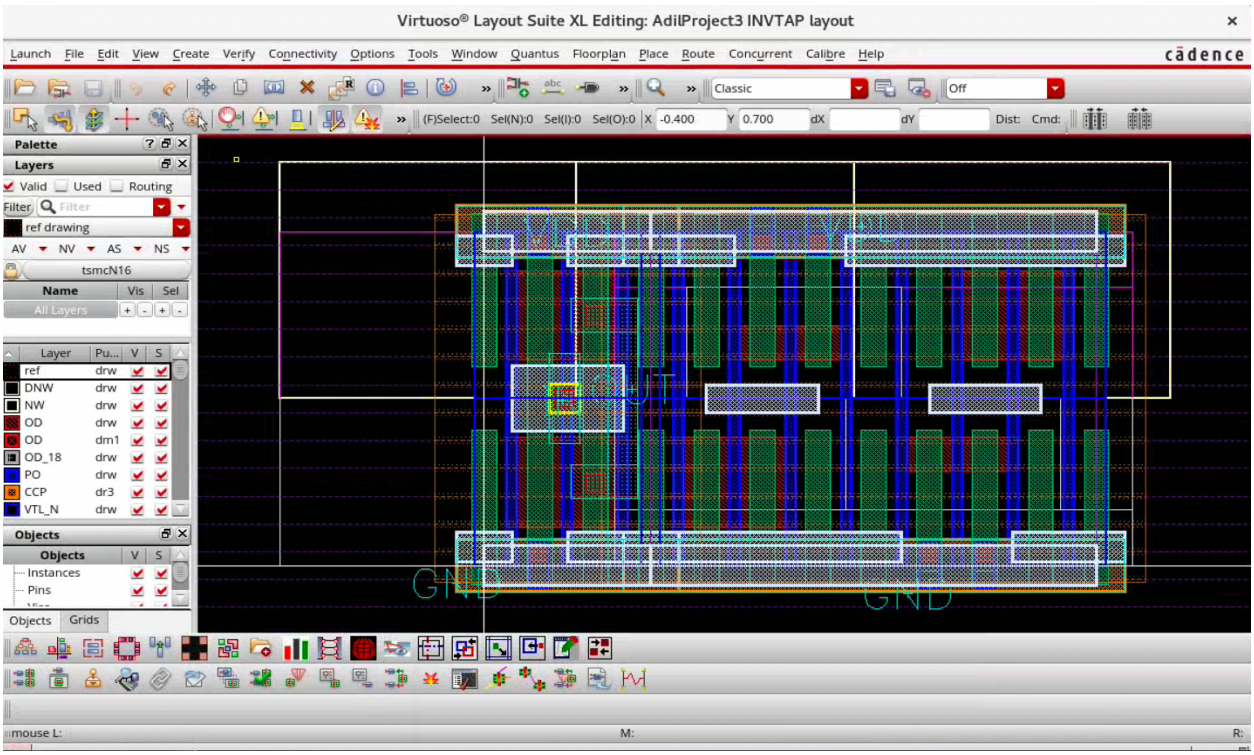


Figure 6: INVTAP layout. INV and TAPCELL abutted with flush prBoundaries, sharing continuous VDD and GND M1 rails. This cell is used for final LVS, PEX, and simulation.

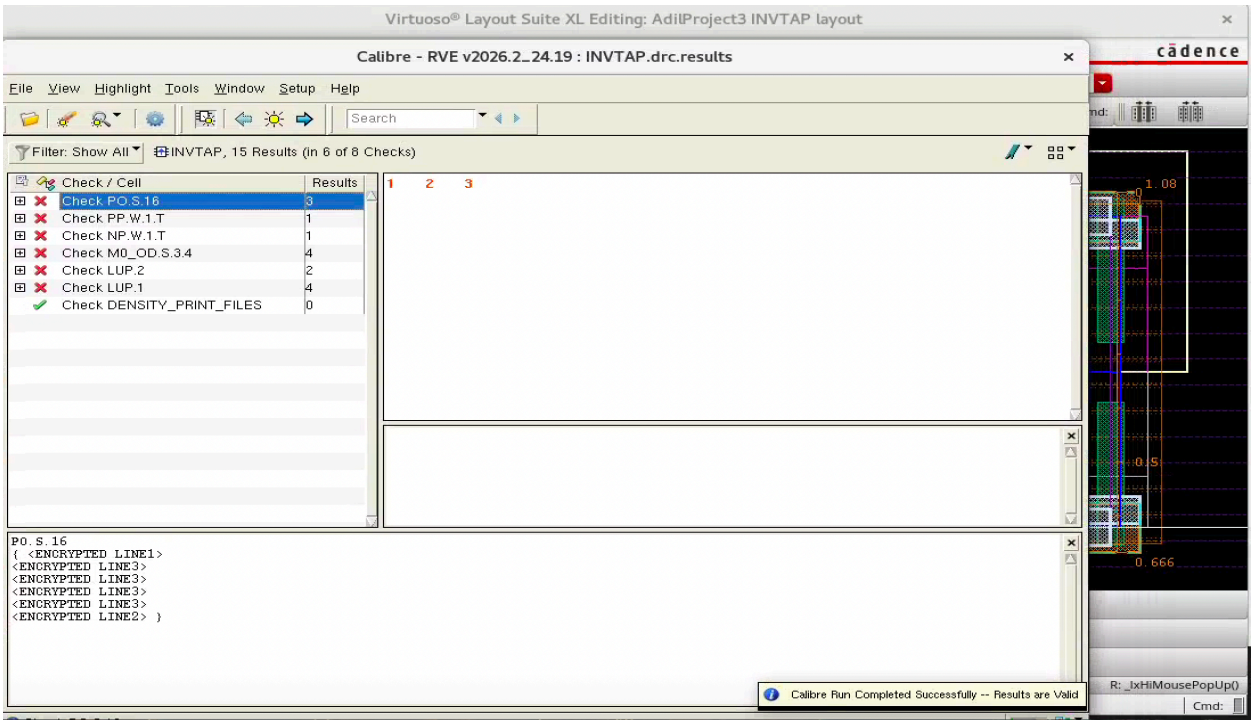


Figure 7: DRC results for the INVTAP cell. After adding the tap cell, PO.R.15 and LUP.6 are gone and the remaining checks are PO.S.16, PP.W.1.T, NP.W.1.T, M0_OD.S.3.4, LUP.2, and LUP.1. These match the tutorial's expected residual set with one additional PP.W.1.T width check, the P-implant counterpart of NP.W.1.T, arising from the same cell abutment.

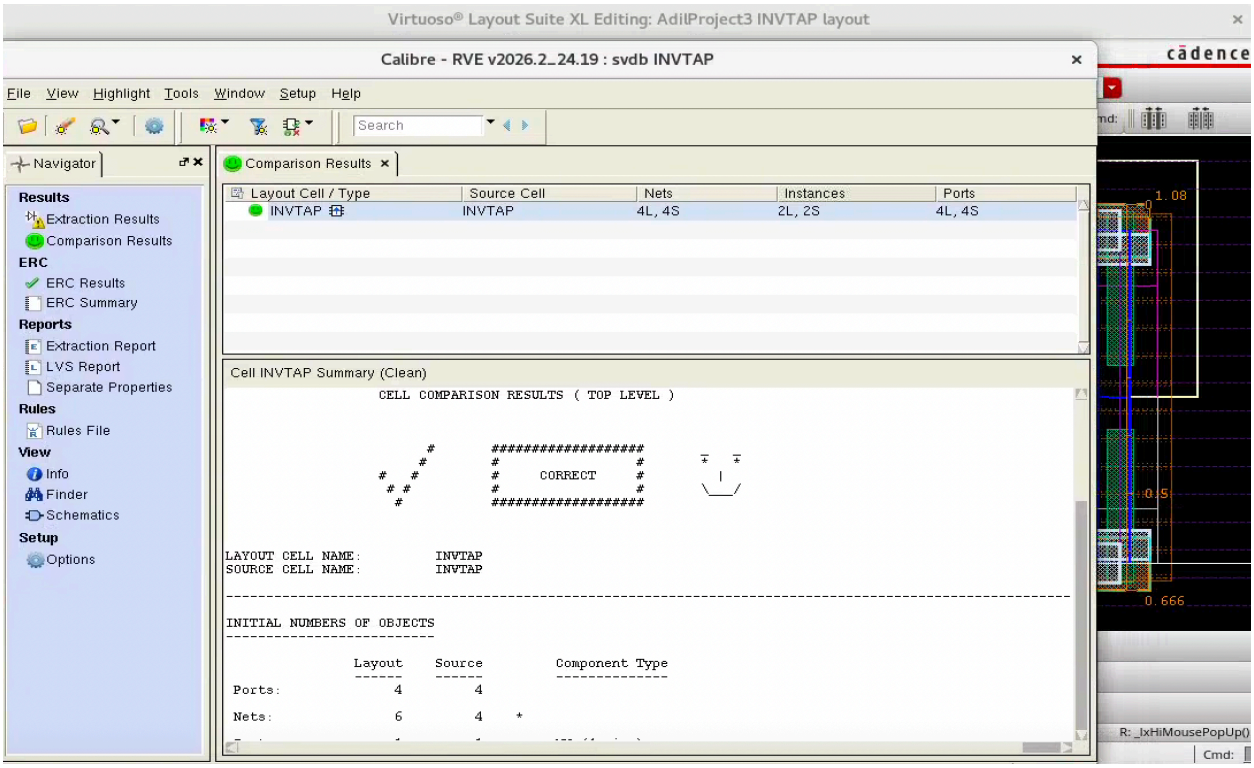


Figure 8: LVS results for the INVTAP cell. Calibre nmLVS reports CORRECT: 4 ports, 4 nets, and 2 instances match between layout and schematic.

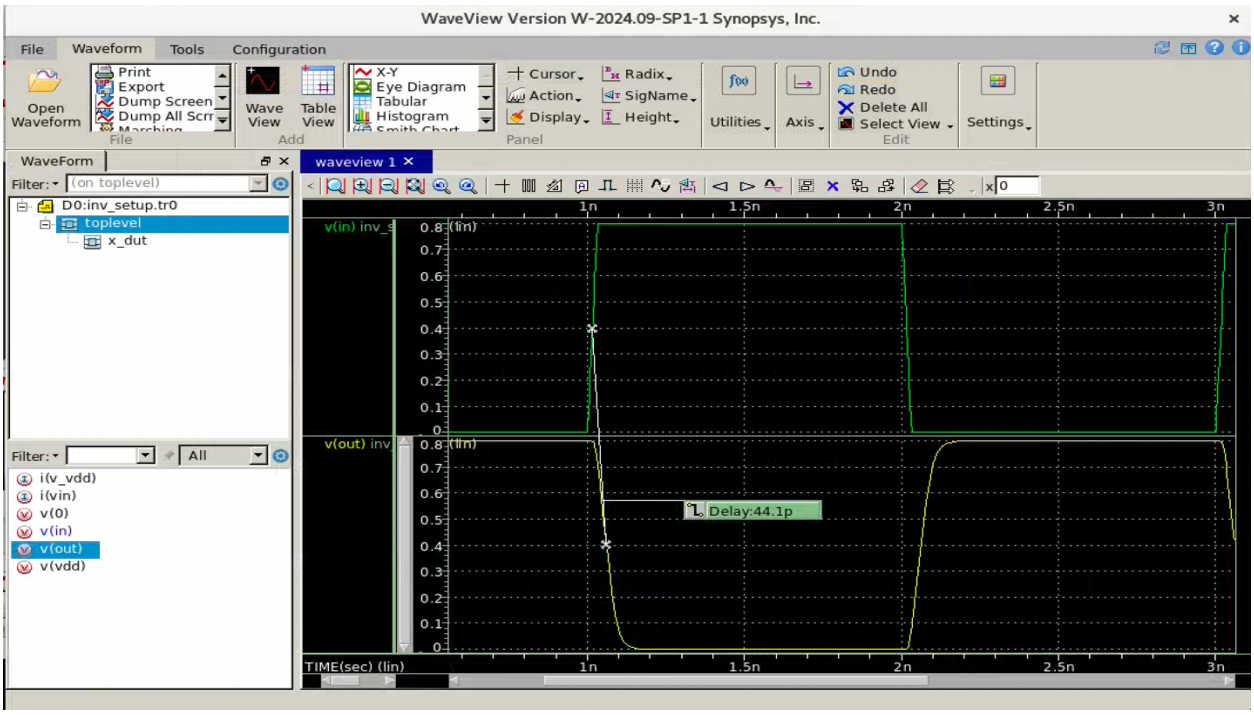


Figure 9: Fall delay (tHL) measurement. Cursors at the 50% point (0.4V) of the rising input and the falling output. Measured tHL = 44.1ps.

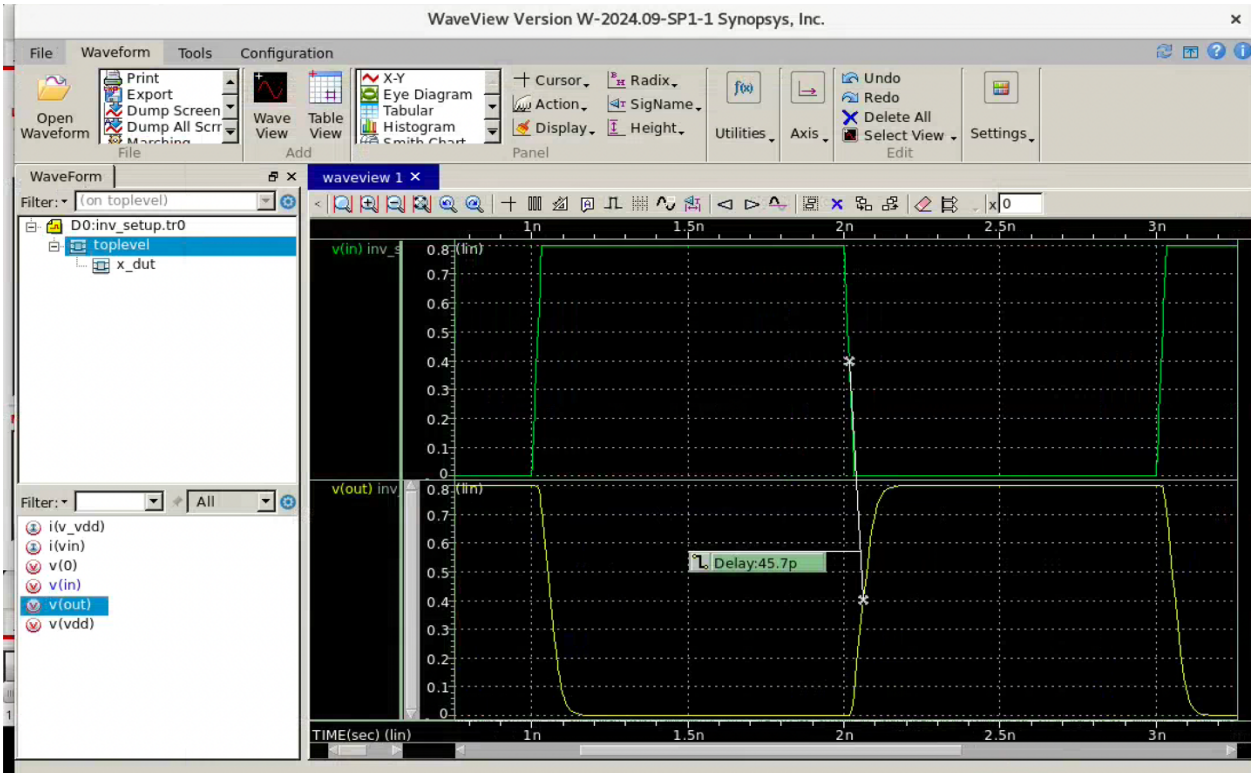


Figure 10: Rise delay (tLH) measurement. Cursors at the 50% point (0.4V) of the falling input and the rising output. Measured tLH = 45.7ps.

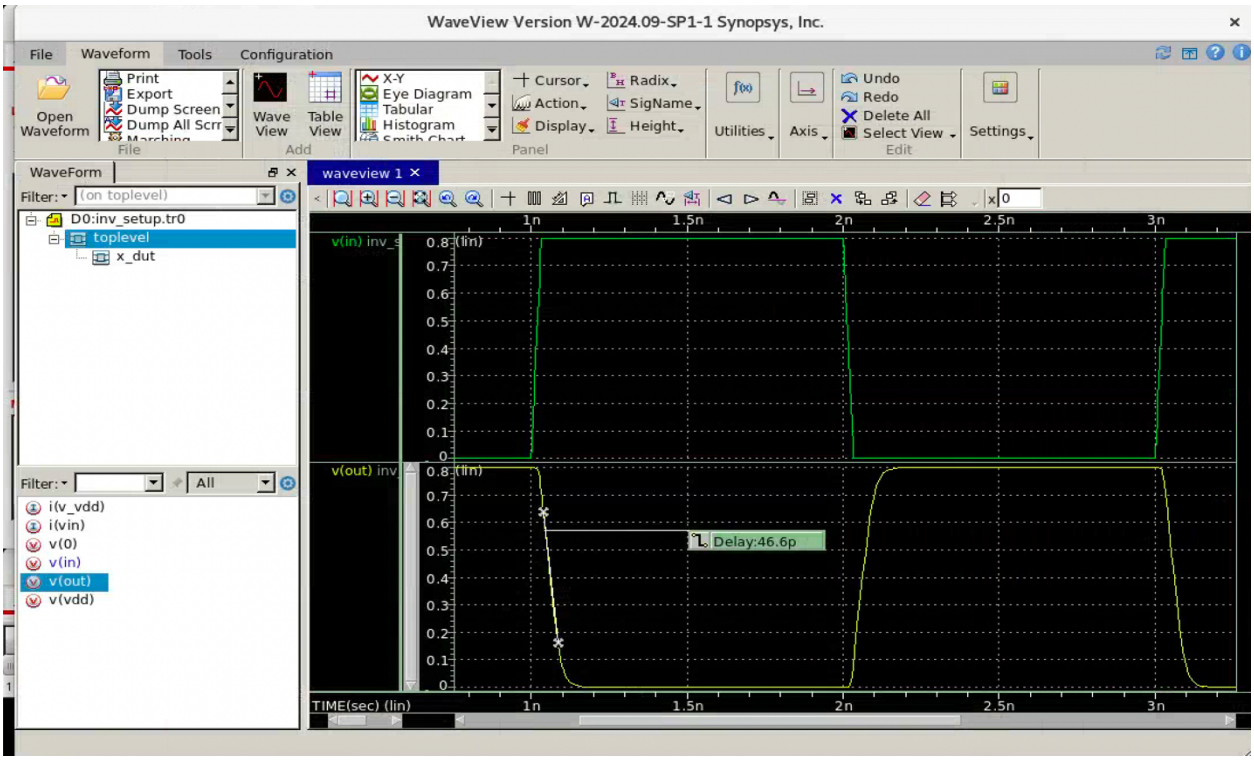


Figure 11: Output fall edge rate (80% to 20%) measurement. Cursors at 0.64V and 0.16V on the falling edge of v(OUT). Measured fall edge rate = 46.6ps.

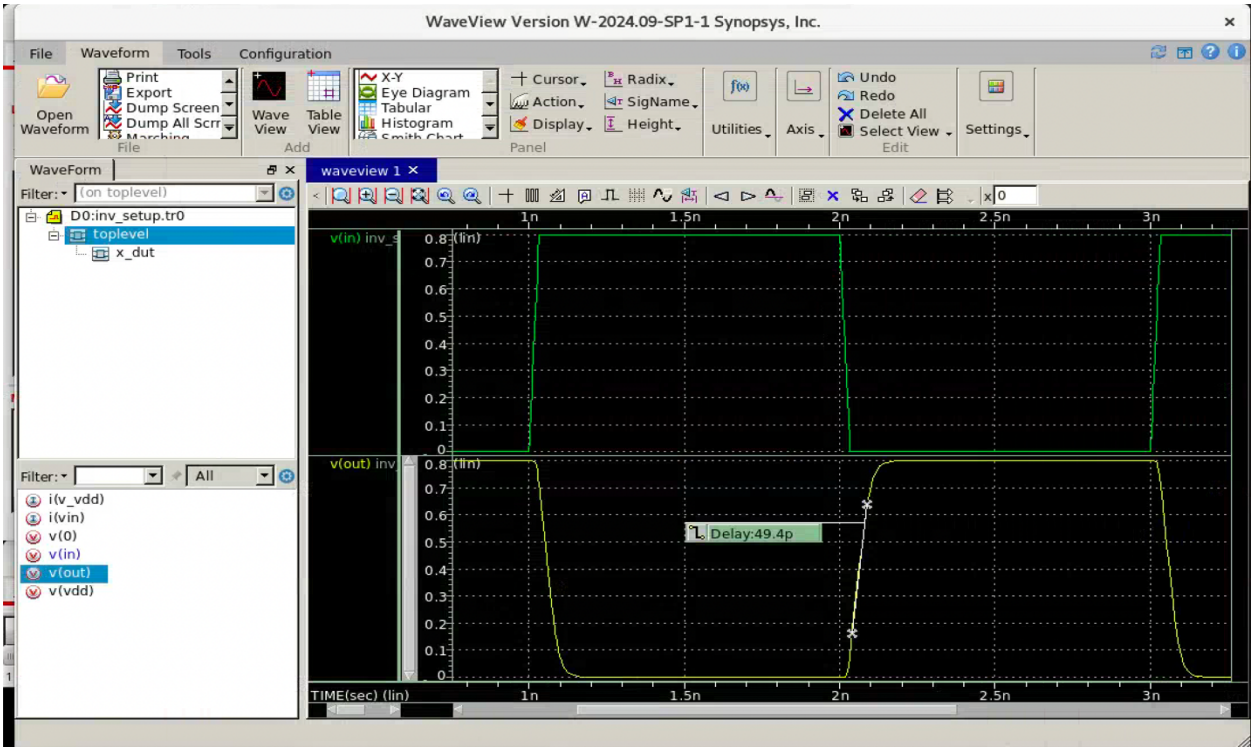


Figure 12: Output rise edge rate (20% to 80%) measurement. Cursors at 0.16V and 0.64V on the rising edge of v(OUT). Measured rise edge rate = 49.4ps.

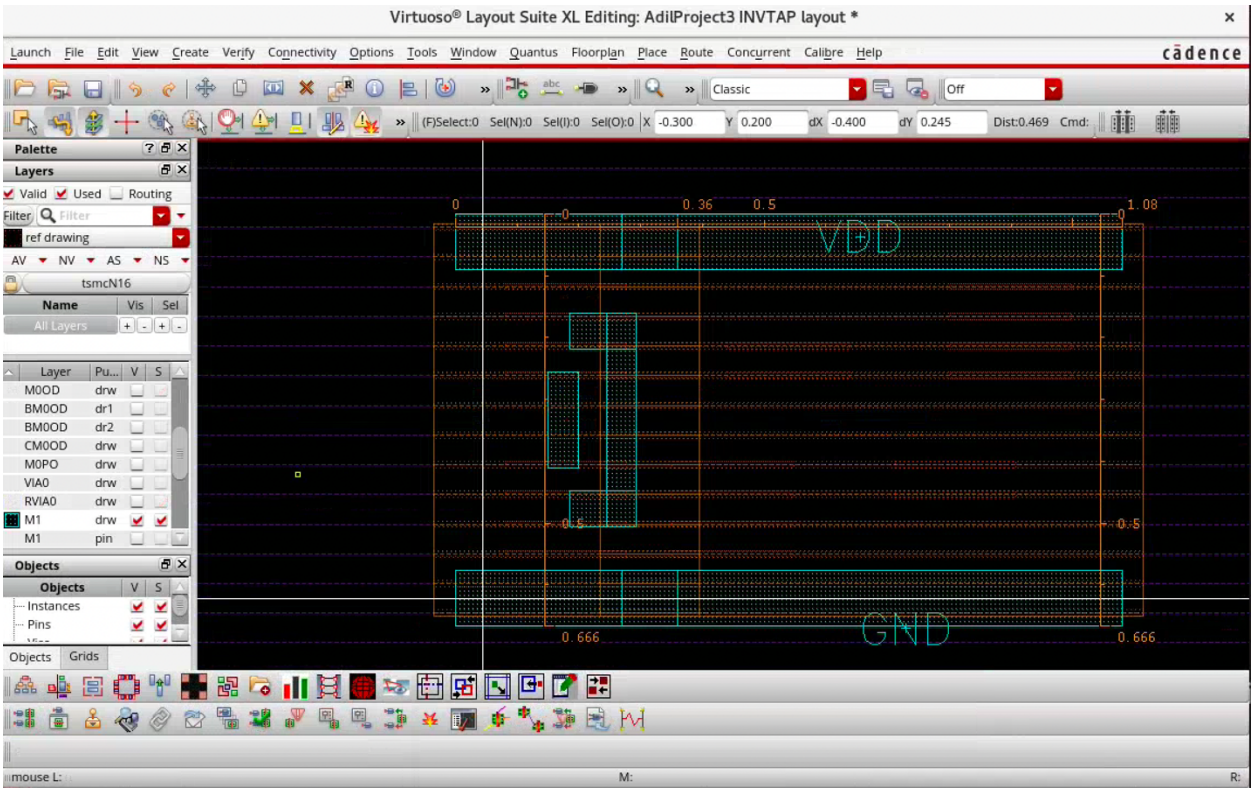


Figure 13: Metal1 bounding-box measurement of the INV cell. Only the M1 layer is displayed. The ruler at 0.36 µm marks the horizontal extent of the inverter's M1 (leftmost to rightmost M1 edge); the vertical ruler spans from the top of the VDD rail to the bottom of the GND rail. Width = 0.36 µm, height = 0.666 µm, area = 0.240 µm^2. The full INVTAP cell including the tap cell measures 1.08 × 0.666 µm = 0.719 µm^2.

Spice Test Setup File:

```
{engnx01a:~/tsmcn16/spice/inv} cat inv_setup.mt0
$DATA1 SOURCE='HSPICE' VERSION='O-2018.09-2 linux64' PARAM_COUNT=0
.TITLE '* transistor models'
 tin_slew       tphl          tplh          trise
 tfall          temper        alter#
  2.000e-11      4.411e-11      4.570e-11      4.936e-11
  4.664e-11       25.0000      1
{engnx01a:~/tsmcn16/spice/inv} cat ~/tsmcn16/spice/inv/inv_setup.sp
* Transistor Models
.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-
001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16a
dfp_spice_model_v1d0.l' FF_svt
.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-
001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16a
dfp_spice_model_v1d0.l' Total_svt
.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-
001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16a
dfp_spice_model_v1d0.l' stat
.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-
001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16a
dfp_spice_model_v1d0_aux.l' FF_mpodesvt
.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-
001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16a
dfp_spice_model_v1d0_aux.l' Total_mpodesvt

* extracted cell
.include 'INVTAP.cbest.pex.sp'

.option POST=2 RUNLVL=5

X_DUT GND OUT VDD IN INVTAP

* Supplies and input (VDD = 0.8 V, 20 ps 20-80% input slew)
V_GND GND 0 0
V_VDD VDD GND 0.8
VIN IN GND pwl(0ps 0v 1000ps 0v 1033.33ps 0.8v 2000ps 0.8v 2033.33ps 0v 3000ps 0v 3033.33ps 0.8v
4000ps 0.8v)

* Load
COUT OUT GND 2e-14

* Transient analysis
.TRAN 1e-12 5e-09

* Measurements
.measure tran tin_slew trig v(in) val=0.16 rise=1 targ v(in) val=0.64 rise=1
.measure tran tphl  trig v(in) val=0.4 rise=1 targ v(out) val=0.4 fall=1
.measure tran tplh  trig v(in) val=0.4 fall=1 targ v(out) val=0.4 rise=1
.measure tran trise trig v(out) val=0.16 rise=1 targ v(out) val=0.64 rise=1
.measure tran tfall trig v(out) val=0.64 fall=1 targ v(out) val=0.16 fall=1
```

.end

Extracted Spice Netlist:

```
*
*
*
*                LINUX        Mon Oct 12 20:34:12 2026
*
*
*
*  PROGRAM  advgen
*
*  Name         : advgen - Quantus - (64-bit)
*  Version      : 20.1.2-p025
*  Build Date    : Thu Sep 3 13:54:09 PDT 2020
*
*  HSPICE LIBRARY
*
*
*

*
.SUBCKT INVTAP GND OUT VDD IN
*
*
*  caps2d version: 12
*
*     SUBCIRCUIT CALLS
*
XX2 GND 21#2 GND npode_svt_mac
+ nfin=4 l=1.6e-08
XX3 OUT 22#3 GND npode_svt_mac
+ nfin=4 l=1.6e-08
XX4 VDD 21 VDD ppode_svt_mac
+ nfin=4 l=1.6e-08
XX5 OUT 22 VDD ppode_svt_mac
+ nfin=4 l=1.6e-08
*
*     TRANSISTOR CARDS
*
*
MM0     OUT     IN      GND     GND     nch_svt_mac     L=1.6e-08       nfin=4
MM1     OUT     IN      VDD     VDD     pch_svt_mac     L=1.6e-08       nfin=4
*
*
*     RESISTOR AND CAP/DIODE CARDS
*
Rt_2_5  OUT     OUT      67.3120        $active
Rt_1_1  VDD     VDD     114.6617        $active
Rt_1_3  VDD     VDD     114.6617        $active
Rt5     OUT     OUT      68.4559        $active
```

```
Rs1     VDD     VDD      10.7329      $M0_STI
Rs2     GND     GND      10.7329      $M0_STI
Rs3     VDD     VDD      21.4521      $M0_STI
Rs4     VDD     VDD      21.4521      $M0_STI
Rr2     IN      IN       17.2243      $poly
Rq_1_1  VDD     VDD      12.4311      $M0_OD
Rq_1_3  VDD     VDD      67.3120      $M0_OD
Rq2     GND     GND      12.9002      $M0_OD
Rq3     GND     GND      68.4559      $M0_OD
Rq7     VDD     VDD       4.8481      $M0_OD
Rq9     VDD     VDD       4.8481      $M0_OD
Ro1     IN      IN        0.2708      $M0_PO
Ro2     IN      IN        0.2708      $M0_PO
Ro3     IN      IN        1.3973      $M0_PO
Ro4     IN      IN        0.2708      $M0_PO
Ro5     IN      IN        0.6824      $M0_PO
Ro6     IN      IN       47.5041      $M0_PO
Ro7     IN      IN       47.5041      $M0_PO
Ro8     IN      IN        4.7543      $M0_PO
Rl1     IN      IN     3.797e-02      $M1
Rl2     OUT     OUT       7.2818      $M1
Rl3     OUT     OUT       7.5957      $M1
Rl4     VDD     VDD       0.3487      $M1
Rl5     VDD     VDD       1.2638      $M1
Rl6     VDD     VDD       0.1743      $M1
Rl9     GND     GND       0.5192      $M1
*
*     CAPACITOR CARDS
*
*
C1      OUT     VDD     1.14951e-18
C2      VDD     12      7.63508e-19
C3      OUT     22      1.43984e-18
C4      31      VDD     1.64833e-17
C5      21      VDD     1.46189e-17
C6      22      VDD     1.06538e-17
C7      9       VDD     1.93117e-17
C8      11      VDD     1.42917e-17
C9      8       21      5.49498e-17
C10     12      VDD     8.18752e-18
C11     9       21      5.49498e-17
C12     15      VDD     8.61986e-18
C13     16      VDD     8.7101e-18
C14     32      VDD     1.64833e-17
C15     22      10      4.22439e-17
C16     6       VDD     1.0514e-17
C17     VDD     6       2.89447e-18
C18     22      11      4.22439e-17
C19     7       VDD     1.33188e-17
C20     19      VDD     1.49998e-17
C21     23      VDD     9.02098e-18
C22     31      VDD     7.42065e-18
C23     32      VDD     6.13826e-18
```

C24	26	VDD	9.06278e-18
C25	27	VDD	8.93623e-18
C26	33	VDD	8.60399e-18
C27	30	VDD	1.33497e-17
C28	VDD	32	1.64833e-17
C29	IN	OUT	1.0089e-17
C30	VDD	26	1.64833e-17
C31	VDD	7	2.91138e-18
C32	IN	VDD	1.46916e-18
C33	10	23	4.234e-17
C34	21	VDD	2.80602e-17
C35	11	23	4.15046e-17
C36	23	3	4.25581e-17
C37	23	12	4.15046e-17
C38	12	31	4.18192e-17
C39	VDD	19	3.45801e-18
C40	21	VDD	2.82427e-18
C41	IN	21	3.92853e-18
C42	VDD	15	7.63508e-19
C43	3	24	4.21612e-17
C44	13	26	4.15046e-17
C45	VDD	23	1.73725e-18
C46	IN	22	2.31239e-18
C47	26	14	4.15046e-17
C48	24	4	4.21612e-17
C49	14	27	4.18192e-17
C50	4	25	4.21612e-17
C51	26	15	4.18192e-17
C52	21	VDD	5.27403e-18
C53	VDD	31	1.55135e-17
C54	15	27	4.15046e-17
C55	25	13	4.234e-17
C56	27	16	4.15046e-17
C57	32	VDD	1.18793e-17
C58	21	GND	5.32426e-18
C59	16	33	4.234e-17
C60	VDD	32	1.32225e-17
C61	33	6	4.6398e-17
C62	9	VDD	1.91854e-18
C63	6	34	4.6398e-17
C64	VDD	26	1.4102e-17
C65	VDD	26	1.45508e-18
C66	17	30	4.15046e-17
C67	34	7	4.6398e-17
C68	VDD	27	2.32734e-19
C69	7	30	4.68868e-17
C70	VDD	27	1.57537e-18
C71	30	18	4.39531e-17
C72	29	17	4.18192e-17
C73	30	19	4.48065e-17
C74	VDD	33	1.57537e-18
C75	VDD	9	9.88715e-20
C76	IN	VDD	3.81699e-19

C77	12	VDD	5.46317e-18
C78	VDD	30	1.86263e-18
C79	VDD	15	2.31514e-18
C80	VDD	15	3.45801e-18
C81	VDD	16	3.45801e-18
C82	31	VDD	6.496e-18
C83	VDD	15	1.31829e-19
C84	IN	VDD	7.04738e-18
C85	OUT	22	1.91845e-17
C86	32	VDD	4.34258e-17
C87	OUT	22	2.07864e-17
C88	VDD	OUT	3.20648e-18
C89	VDD	12	1.31829e-19
C90	32	VDD	6.496e-18
C91	22	OUT	2.80602e-17
C92	OUT	VDD	3.1977e-18
C93	OUT	VDD	1.19094e-17
C94	VDD	IN	2.71791e-18
C95	VDD	32	6.496e-18
C96	OUT	GND	3.3774e-18
C97	21	GND	3.93589e-17
C98	22	OUT	2.40392e-17
C99	GND	OUT	8.14676e-18
C100	IN	OUT	4.00927e-18
C101	VDD	26	6.75906e-18
C102	GND	IN	3.93589e-17
C103	26	VDD	2.19386e-17
C104	IN	VDD	2.80602e-17
C105	GND	IN	5.14657e-18
C106	21	GND	4.92169e-19
C107	OUT	IN	2.12637e-17
C108	IN	21	4.41891e-18
C109	21	VDD	1.64684e-17
C110	21	IN	3.70351e-18
C111	VDD	IN	1.64762e-17
C112	OUT	IN	2.3601e-18
C113	IN	OUT	2.80602e-17
C114	28	GND	1.06634e-16
C115	OUT	VDD	9.42691e-18
C116	IN	GND	5.08091e-19
C117	GND	21	5.15954e-18
C118	GND	IN	5.63219e-19
C119	IN	GND	2.75139e-18
C120	GND	IN	4.22843e-18
C121	31	VDD	2.17129e-17
C122	OUT	IN	2.62102e-19
C123	IN	22	4.45029e-18
C124	OUT	IN	7.25585e-19
C125	VDD	IN	5.09914e-18
C126	IN	22	3.6322e-18
C127	GND	OUT	2.25193e-18
C128	VDD	9	1.20843e-19
C129	IN	OUT	2.40392e-17

C130	IN	GND	1.0798e-18
C131	GND	OUT	4.82403e-19
C132	29	GND	5.63337e-17
C133	27	GND	5.67284e-17
C134	IN	VDD	3.43844e-18
C135	IN	OUT	1.92263e-17
C136	OUT	VDD	1.22935e-19
C137	OUT	VDD	1.50976e-19
C138	OUT	VDD	1.79702e-19
C139	VDD	OUT	2.29372e-19
C140	VDD	22	1.37321e-19
C141	OUT	GND	5.81753e-18
C142	VDD	GND	6.7648e-18
C143	21	GND	1.22204e-17
C144	22	GND	1.85524e-17
C145	8	GND	2.32985e-17
C146	9	GND	1.57219e-18
C147	10	GND	2.15077e-17
C148	11	GND	8.65212e-18
C149	3	GND	2.57068e-17
C150	12	GND	5.69971e-18
C151	4	GND	2.42165e-17
C152	13	GND	1.91205e-17
C153	14	GND	2.0066e-17
C154	15	GND	4.7788e-18
C155	16	GND	6.89906e-18
C156	6	GND	1.15421e-17
C157	17	GND	1.99567e-17
C158	7	GND	1.05074e-17
C159	18	GND	2.25563e-17
C160	19	GND	2.87318e-18
C161	23	GND	2.50329e-17
C162	31	GND	5.18119e-18
C163	32	GND	4.42331e-18
C164	26	GND	2.04956e-17
C165	27	GND	1.86639e-17
C166	33	GND	4.4256e-18
C167	34	GND	1.1091e-17
C168	30	GND	1.69966e-17
C169	24	GND	1.11888e-17
C170	25	GND	1.48457e-17
C171	28	GND	1.04119e-17
C172	29	GND	1.23647e-17
C173	OUT	GND	9.96446e-20
C174	OUT	GND	1.22167e-17
C175	VDD	GND	2.37396e-18
C176	VDD	GND	5.21064e-19
C177	VDD	GND	5.31367e-19
C178	IN	GND	6.0744e-19
C179	IN	GND	6.0619e-18
C180	VDD	GND	7.99495e-18
C181	VDD	GND	9.54122e-20
C182	IN	GND	1.37753e-18

```
C183    OUT     GND     6.9003e-18
C184    OUT     GND     1.00714e-17
C185    VDD     GND     2.44146e-18
C186    VDD     GND     9.88715e-20
C187    VDD     GND     3.70209e-19
*
*
.ENDS INVTAP
*
```